

T115 Synthetic Data Validation Documentation

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Introduction

This documentation will go over:

- Details of each validation process including the methods or processes used for validation, what information is included in the results, and an analysis of the outcomes.
- Limitations of the synthetic data or any methods used in data generation or validation

Relationship between Student Attrition and Categories

- Get attrition rate vs each variable and draw bar plot for visualisation (please see **Synthetic Dataset Attrition Rates by Categories.xlsx** for the results). For each variable, there is a pooled version and a “Per Year” version. The purpose of this is to visualise the correlations between the variables in the synthetic data and student attrition rate.
- The “Per Year” sheets include two versions of the same table of information: the one on the top-left is in a SQL format and the other that is below the plot is in a pivot table format to produce the plot. Seeing the data in the pivot table format may make it easier to compare attrition rates across different categories in the same year.
- The following variables were considered most relevant to student attrition and thus, attrition rate was gathered for each:
 - Attendance mode
 - Course Admission Load Category
 - Course Level
 - Broad Field of Education
 - Age band
 - Student Gender
 - Socioeconomic Status

- Regional Remote Status
 - First Nations
 - Domestic vs International Students
- The following attrition rates were produced by fitting a beta distribution. Attrition vs:
 - Attendance mode
 - Course Admission Load Category
 - Course level
 - Broad Field of Education
- All other attrition rates were randomly generated – which means statistical trends will likely not follow those of real student data.
- The distribution of attrition rates in the variables where a beta distribution was applied is quite similar to the distribution of randomly generated data. This may mean something went wrong when applying the beta distribution during the synthetic data generation process.

Correlation between Student Attrition and Numerical Variables

- Get attrition rate vs each numerical variable and draw a scatter plot with a linear trendline for visualisation (please see **Synthetic Dataset Attrition Rates by Categories.xlsx** for the results). This is to identify any numerical relationships that may exist between variables in the synthetic data and attrition rates.
- The following numerical variables were considered most relevant to student attrition and thus, attrition rate was gathered for each:
 - EFTSL
 - Cumulative credit points enrolled
 - Cumulative credit points failed
 - Cumulative credit points passed
 - Cumulative credit points withdrawn
 - Age
- All attrition rates were randomly generated – which means statistical trends will likely not follow those of real student data.
- Most plots have an extremely low gradient and R-squared value – indicating that the distributions do not have a linear relationship.
- The plot for attrition rate by EFTSL does not have an unreasonably low gradient but the R-squared value is still low and the scatter plot shows that the distribution of attrition rates is mainly flat with heavy clustering near the low end of EFTSL – revealing that there is no linear relationship for attrition rate and EFTSL.

Reproducing Distributions Provided by Industry Partners

- Reproduce key distributions provided by industry partners to observe whether counts in synthetic data are reasonable (please see **Synthetic Dataset Distributions vs Provided Distributions.xlsx**). The purpose of this is to check that the distributions provided by the industry partners were maintained in the synthetic data.
- The distributions in v2 and v3 were considered most relevant to the use case and thus, their distributions were recreated using the synthetic data:
 - Commencing_vs_continuing_by_foe_pooled
 - Commencing_vs_continuing_by_teaching_period_pooled
 - Cumulative_credit_points_enrolled_by_field_of_education_pooled
 - Cumulative_credit_points_failed_by_field_of_education_pooled
 - Cumulative_credit_points_passed_by_field_of_education_pooled
 - Cumulative_credit_points_withdrawn_by_field_of_education_pooled
 - EFTSL_distribution_by_teaching_period_per_foe_2023
 - EFTSL_distribution_by_teaching_period_per_foe_pooled
 - attendance_mode_pooled_2021-2024_byEnrolments
 - course_admission_load_category_pooled_2021-2024
- The distributions that were gathered by course could not be compared since the course keys in the synthetic data are randomly generated and thus, completely different to the course keys in the real data.
- Course admission attendance mode and course admission study mode are variables that were not included in the synthetic data generation. Therefore, distributions relating to these variables were not produced.
- When generating the synthetic data, the number of rows in the synthetic dataset was set to be half of the number of rows in the real dataset due to computational limitations (the real data has approximately two million rows which our personal devices may not have the computational capacity for producing them). Therefore, it would be reasonable to measure the closeness of the synthetic dataset distributions to the real dataset by the number of rows in the synthetic dataset being approximately half of those in the real dataset. However, a key detail of the synthetic data is that each row has a unique student identifier – which means the synthetic data may have less records but more students. Therefore, distributions using student counts instead of enrolment counts may not follow the above logic.
- The results show that Commencing_vs_continuing_by_foe_pooled mostly follows the expected logic of having half as many counts in the synthetic data than in the real data.
- All other distributions either have random counts in the synthetic data that cannot be attributed to any trend or correlation to the real data, or the synthetic data includes a disproportionate number of categories that have no records in the real data (this is most clearly seen in EFTSL_distribution_by_teaching_period_per_foe_2023 and EFTSL_distribution_by_teaching_period_per_foe_pooled)
- Notes specific to each distribution are included in the Excel sheets.

Limitations of Synthetic Data

- Each row of the student information table has a unique student deidentified hash – which means that the synthetic data does not depict historical data. Thus, the synthetic data is deemed cross-sectional, and the resulting machine learning model should be considered a proof of concept rather than a longitudinal attrition model.
- In the teaching periods table, teaching periods with the same year and teaching period name had different dates. This is not a realistic depiction of teaching period dates. However, there are multiple teaching periods with the same teaching_period string and calendar_year that have different keys where the only difference is the start and end dates. Therefore, the data will remain unchanged – leaving teaching periods with the same name and calendar year to have different dates – but this may be revised in the future.

Feature Selection Decisions

The following are feature selection decisions that arose or were discovered during the validation process. This is not a conclusive list of feature selection decisions as more motivations for feature selection may arise in other processes.

- In the student information table, the variable “is_commencing” and its variants represent the same information as “commencing_continuing” and its variants in the synthetic dataset. To remove redundant variables when training, validating, and testing the machine learning model, only “commencing_continuing” and its variants will be used.
- In the student information table, the variable “is_international_student” represents the same information as “international_domestic_enrolment”, “international_domestic_student” and “student_is_international_student” in the synthetic dataset. To remove redundant variables when training, validating, and testing the machine learning model, only “international_domestic_student” will be used.
- In the course information table, the variable “course_level” represents the same information as “course_group” in the synthetic dataset. To remove redundant variables when training, validating, and testing the machine learning model, only “course_level” will be used.